In this page, we follow the book of Lars-Ahlfors.

Def. Let an arc γ be given by equation $z = z(t)$ $\alpha \le t \le \beta$ in a region Ω.
Let f be a continuous map which values defined in Ω.
An arc γ' which is given by $w = f(z(t))$ is called the image of γ

Consider the f is analytic and $z(t)$ is smooth. Then,

$$w'(t) = f'(z(t)) \cdot z'(t)$$

For t_0, if $z'(t_0) \neq 0$ and $f'(z(t_0)) \neq 0$, then $w'(t_0) \neq 0$, Now,

$$\arg w'(t_0) = \arg f'(z_0) + \arg z'(t_0).$$

This equation means the difference of the tangent angle of the curve γ between the image γ' is exactly $\arg f'(z_0)$, regardless of the curve.

[Thm.
Suppose that $f(z)$ is analytic at z_0 with $f'(z_0) \neq 0$.
Let $C_1: z_1(t)$ and $C_2: z_2(t)$ be smooth curves in the domain
that intersect at $z_0 = z_1(t_0) = z_2(t_0)$, with $C_1': w_1(t)$ and $C_2': w_2(t)$,
the images of C_1 and C_2 respectively.
Then, the angle between C_1 and C_2 measured from C_1 to C_2 is equal to
the angle between C_1' and C_2' measures from C_1' to C_2',